Motor Calculation

Car specifications:

Specification	Details
Mass	280Kg (Inc. driver)
Frontal Area	1.0735m2
Distance to Travel	22km
Time for Endurance Test	30 mins

Wheel Requirements:

Requirement	Mean	Peak
Power	12kW	40kW
Wheel speed	850 rpm	1650 rpm
Torque	132.16 Nm	227 Nm
Energy	4.224 kWh	7.376 kWh

Using gear ratio of 3:1 as a chain drive gear system

The motor requirement comes out as:

Requirement	**Mean**	**Peak**
Wheel speed	2550 rpm	4950 rpm
Torque	44.05 Nm	75.6 Nm
Power	12kW	40kW

Motenergy ME1905 (ME1507) Brushless 14.5kW Cont. 44kW Pk. IPM Motor 48-120V

Specification	**Mean**	**Peak**
Power	14.5kW	44kW
Rotor Speed		6000 RPM
Current	220A AC	550A AC for 1 min
Torque	41.8 Nm	104.5 Nm
Voltage (DC)	48V Min	120V max

Controller Selection

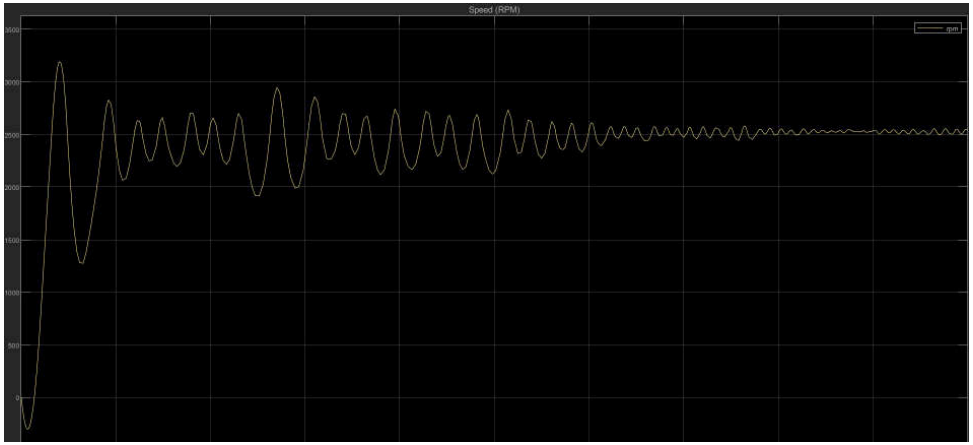
For the above specified motor, the controller that fits the requirements is:
SEVCON Gen4 Size-6 80V, specifications of which are tabularized as :

Model	Size 6
Nominal Battery Voltage	96 – 120 VDC
Maximum Operating Voltage	150 VDC
Minimum Operating Voltage	48 VDC
Peak Phase Current (2 min)	550 A
Boost Phase Current (10 sec)	660 A
Continuous Phase Current (60 min)	220 A

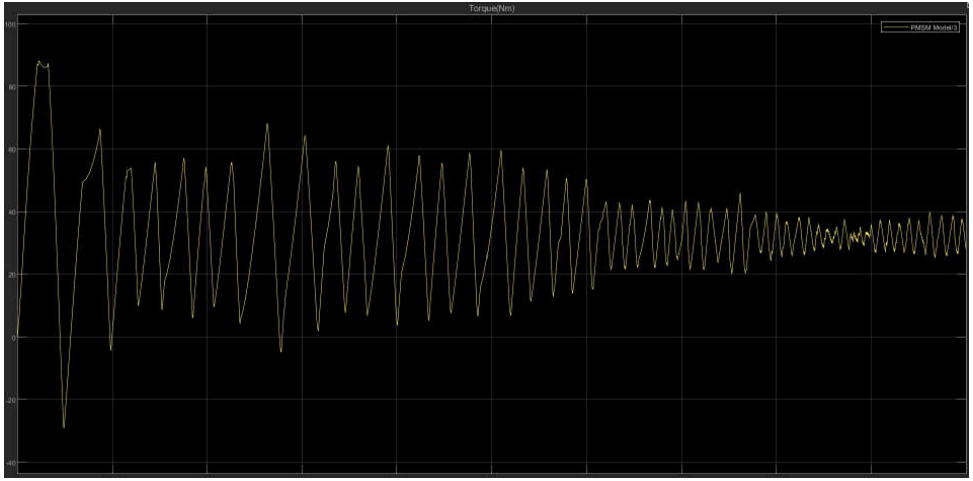
Motor Driver Simulation in Matlab Simulink:

Parameter	Value
Vdc	120 V
Tm	32 N-m
Reference Speed	2500 RPM

At 0.025 sec a load torque of 32 Nm is applied to the motor, which causes a slight decrease in the speed. Initially the commanded torque is equal to maximum capability of the motor and when loaded the electrical torque increases to satisfy the desired load torque.



Reference Speed 2500 rpm is maintained



Torque is modulated to maintain the speed

Motor/Controller Thermal Calculations

Mathematical Data:

Parameter	Value
Mass of the Motor (m)	21.4 kg
Specific Heat Capacity (c)	500 J/kg·°C
Temperature Difference (ΔT)	115°C (140°C - 25°C)
Surface Area of the Motor (A)	0.7035 m^2
Heat Transfer Coefficient (h)	500 W/m^2·°C (for Water Cooling)

The Calculation data for the motor thermal management comes out as:

Calculation	Result
Heat Energy to be Dissipated (Q)	1,230,500 J
Heat Transfer Rate (Q)	40,250 W
Cooling Time (t)	30.571 s

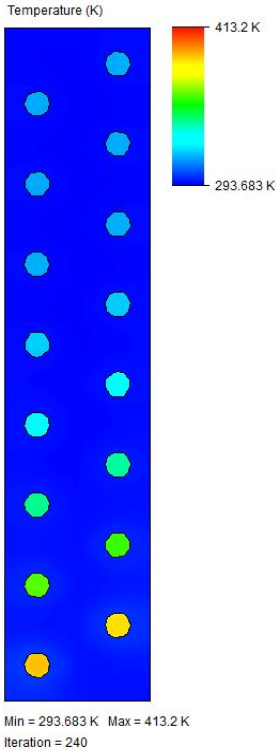
Simulation Verification:

Calculation Parameters:

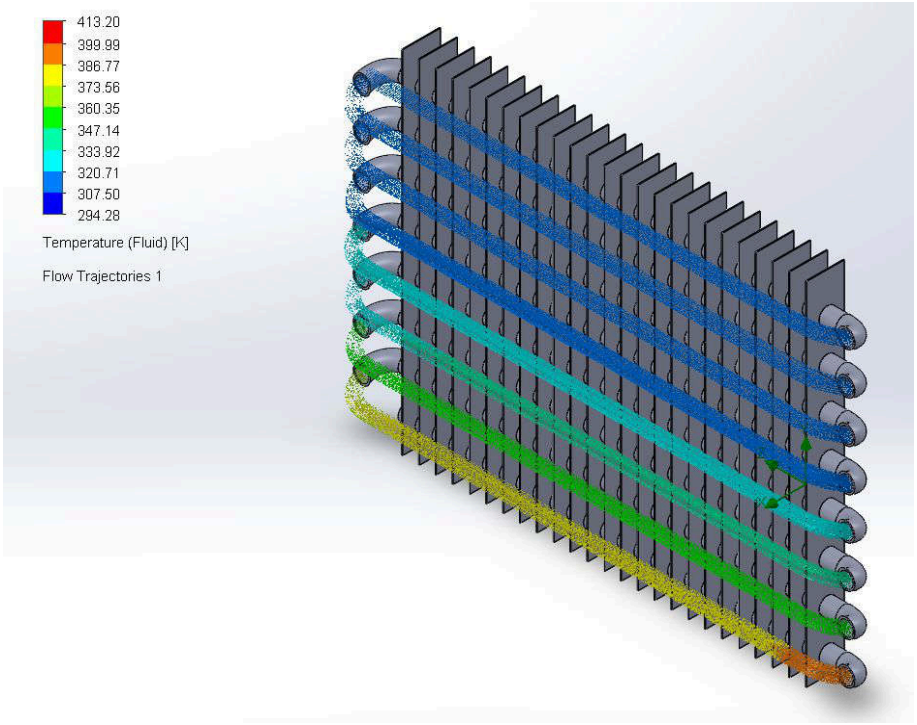
Parameters	Value
Temperature of liquid	423 K (inlet)
Liquid type	Water
Wall heat conduction coefficient	500 W/m2/k
Outlet temperature	292K

Design Dimensions:

Parameter	Value
Length	20 cm
Width	30 cm
Dia of pipe	9 mm



Side view of radiator



Demonstration of the fluid temperature inside radiator

Sensor and Electronics

Brake Pedal Sensor

For the Brake pedal sensor we will be using Honeywell MLH03KPSL01A sensor.

Parameters	Range
Supply Voltage	6V to 24V
Output Voltage	0.5V to 5V
Operating Temperature	-40 C to 125 C
Pressure range	0 to 3000 psi

Current Sensor

For the Current sensor we will be using LEM CAB-SF 500-C current sensor.

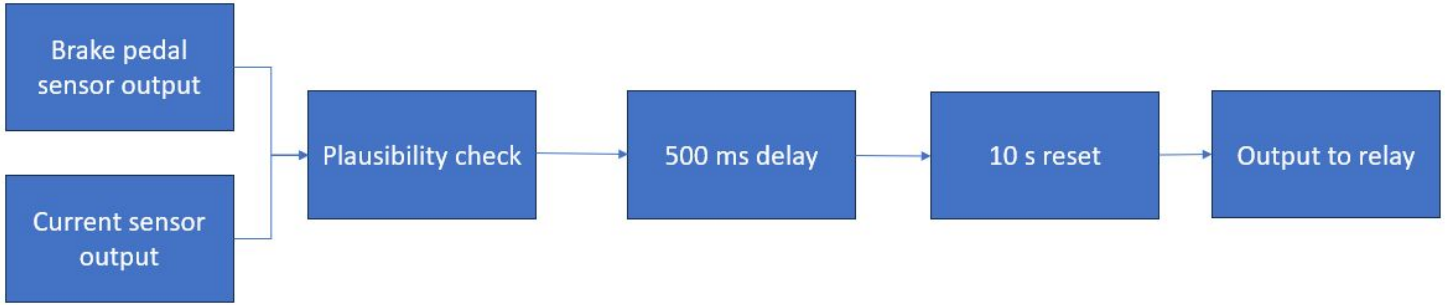
S no.	Specification	Range
1.	Supply Voltage	8.5V to 16V
2.	Output Type	CAN signal
3.	Operating Temperature	-40 C to 85 C
4.	Nominal current	500 A

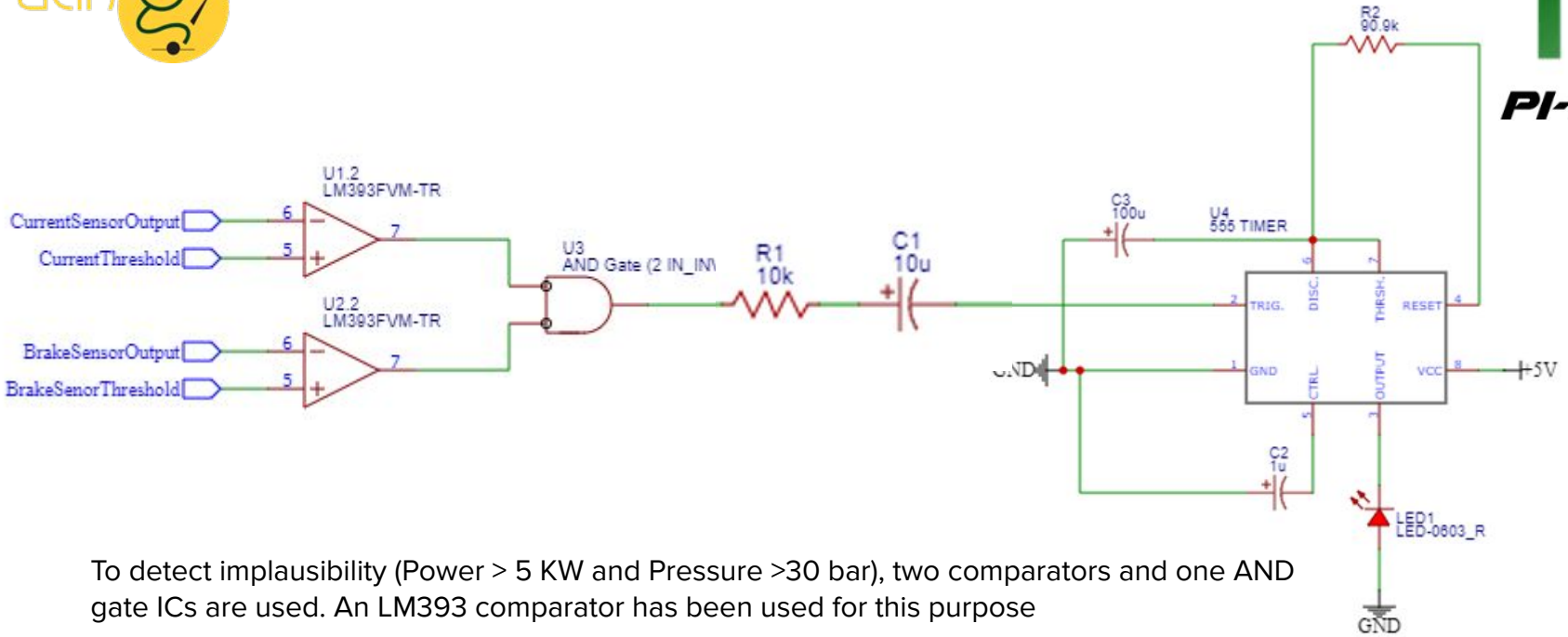
Brake System Plausibility Device:

A DC Current sensor LEM CAB-SF 500-C

The Brake pressure sensor Honeywell MLH03KPSL01A is used for appropriate rating.

- Power Delivered to the motor >= 5KW,
- Hard Braking, if the pressure in brake pedal line > 30 bar.





To detect implausibility (Power > 5 KW and Pressure >30 bar), two comparators and one AND gate ICs are used. An LM393 comparator has been used for this purpose

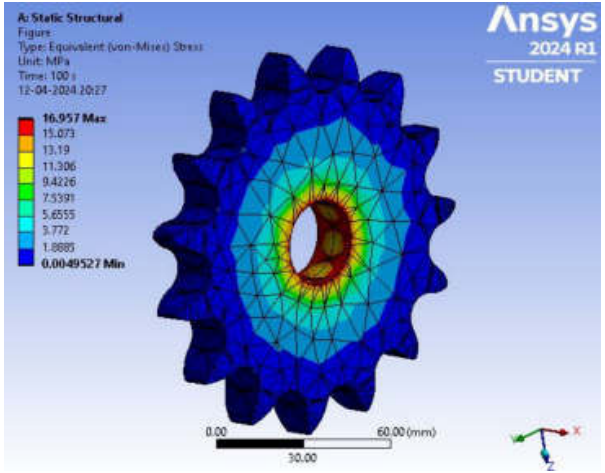
R = 10 kΩ and C = 10uF we can get 99% of charging and the required 500 ms of delay.

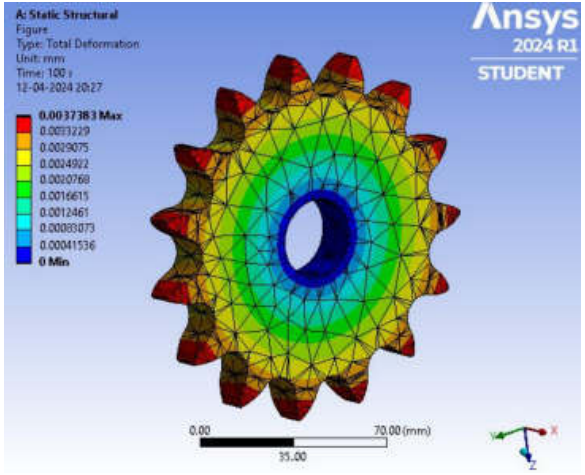
The time delay with a monostable mode is calculated as

Taking C = 100uF and R =90.9 kΩ, we get time delay as
T = 1.1 * 0.0001 * 90900
T = 9.999 sec ≃ 10 sec

Differential and Drivetrain

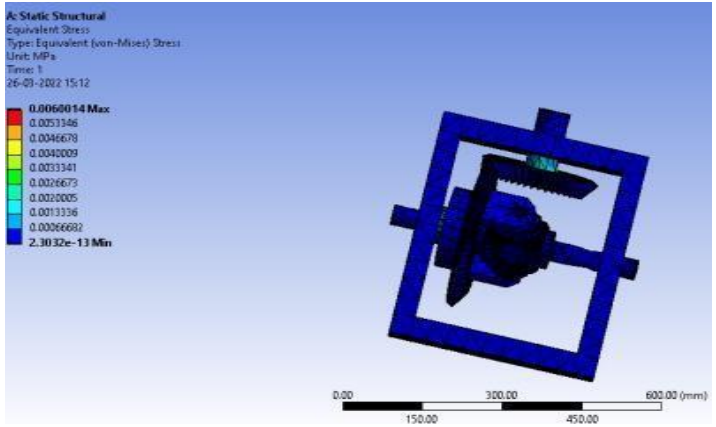
To meet the goals of our vehicle, we used a gear ratio of 3:1 as a chain drive gear system
Chain and sprocket with open differential is selected for power transmission from Motor To wheels.

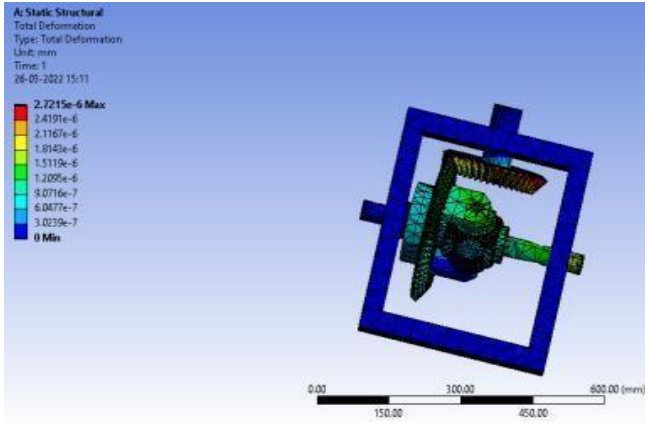




This gearing arrangement makes up the open type differential, and is the most common type of differential.

Moment (N-mm)	Material	Von Mises Stress (MPa)	Total Deformation (mm)
100	Structural Steel	5.81E-05	0.00588
500		2.90E-05	0.029399
1000		5.81E-06	0.058798
1500		8.71E-05	0.088197

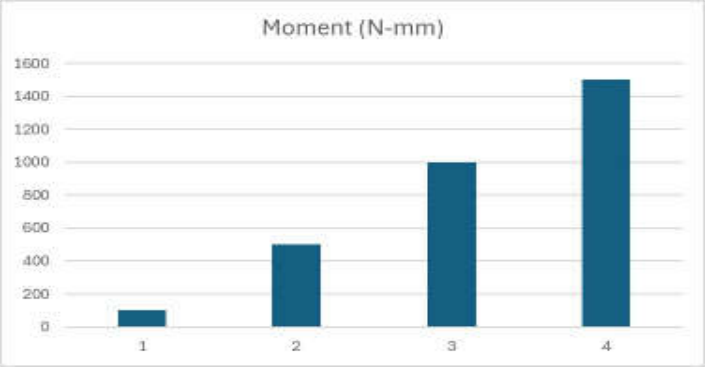


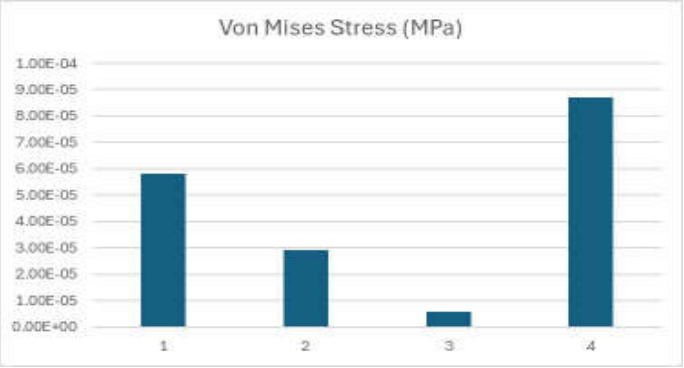


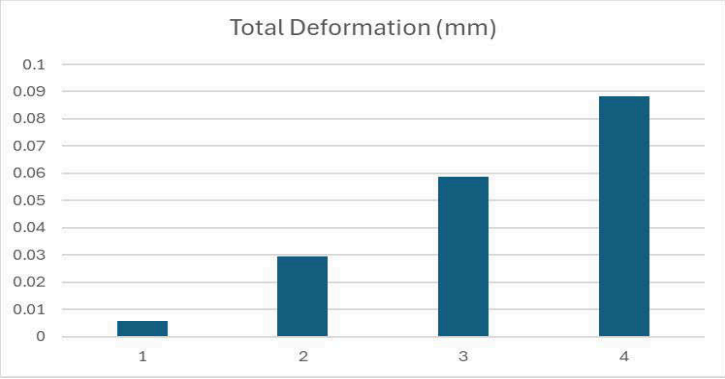
IIITDMJ racing

PI-EV

Graphical analysed data for open differential :







Cell Selection
&
Energy and Power Estimations

The points highlighting the technical advantages of using **3.6V 50Ah NMC prismatic cells for a 200Ah, 120V accumulator pack** :

Specification	Value
Nominal Voltage	3.6V
Capacity	50Ah
Energy Density	200-250 Wh/kg
Thermal Stability	Up to 60°C (operating)
Cycle Life	2000-3000 cycles
Charge Efficiency	95-98%
Discharge Efficiency	95-98%
Balancing	Active/Passive balancing
Safety Features	Overcharge protection, Over-discharge protection, Short-circuit protection, Thermal management systems
Cost per kWh	$150-250 per kWh
Mechanical Stability	High mechanical strength, Robust prismatic casing

For Effective run time, ***Teff=H(C/(IH)) ^k***
Using NMC Cell Of 50Ah
k~0.9
Power = 44kW
Capacity = 200Ah
Continuous discharge rate, H = 2C
Voltage = 120V
Current (P/V) =366.6 A ~ 380 A or 400A
Effective run time (t=H(C/ (IH)) ^k) =34.46 min

Parameter	Value
Peak Voltage	153V
Nominal Voltage	131.4V
Peak Current	340A @1.7C
Continuous Current	100A @0.5C

Calculating Power and Energy by using the formulas:

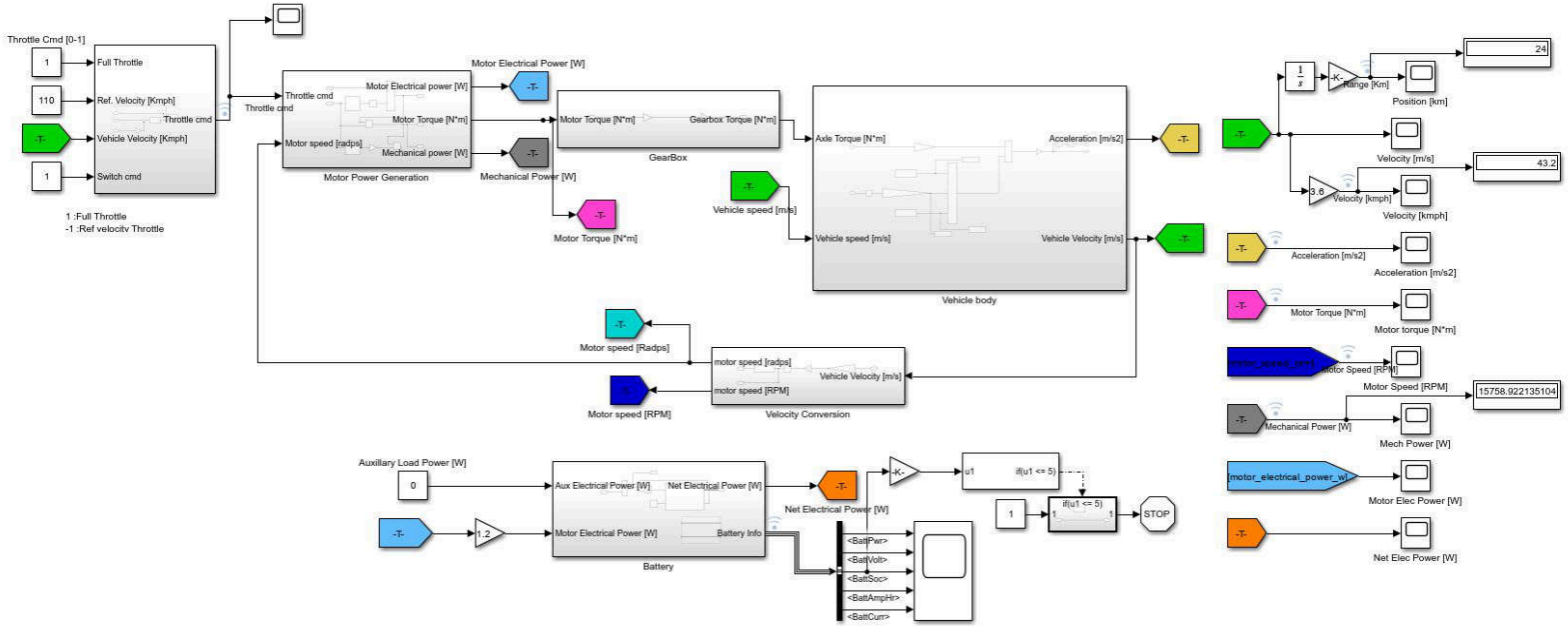
- Power = Voltage x Current
- Energy = Effective capacity x Nominal Voltage per cell x Cells in series

Parameters	Value
Peak Power	52.02kW
Continuous Power	13.14kW
Energy at 1.7C	26.28 kWh

Next slides will be showing the Powertrain simulation of the data of the battery pack, this model was created with the help of Simulink.

Validation using Matlab

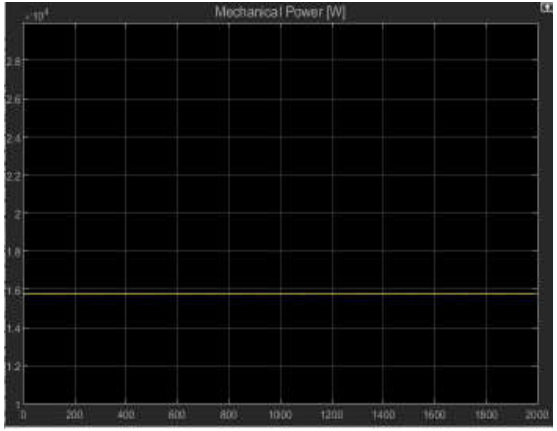
Simulation of EV Powertrain model



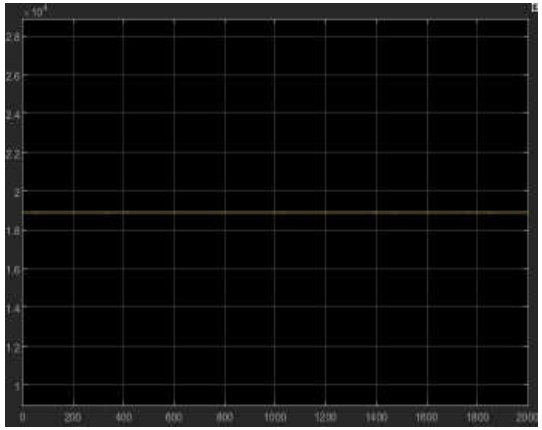
*the model simulated for 1600s

Input Parameters for the above model are tabularized below:

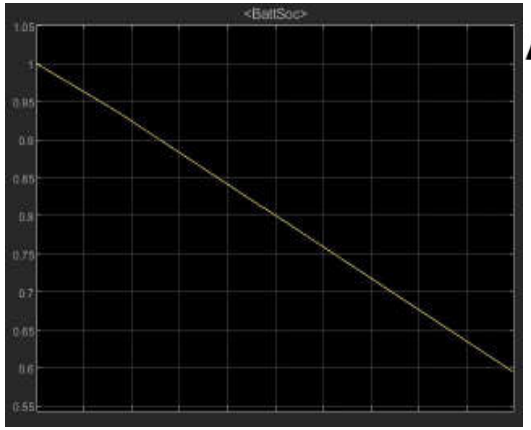
Specification	Details
Mass	280Kg (Inc. driver)
Frontal Area	1.0735m2
Time	30 mins
Motor speed	2500-6000 rpm
Torque	30-70 Nm
Cell Configuration	50Ah 3.65V NMC Prismatic cells
Arrangement	4P 36S



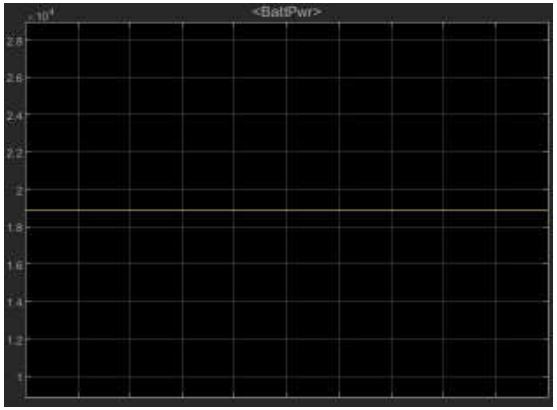
(a)Mechanical Power



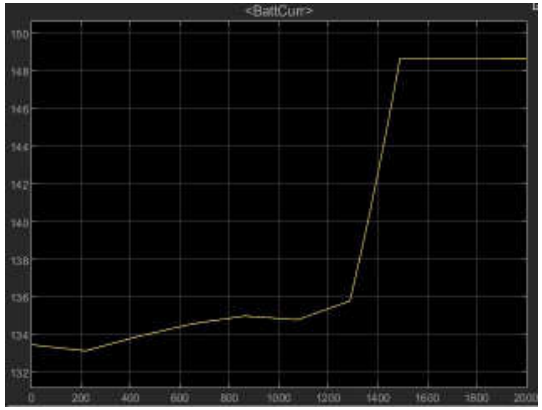
(b)Net Electrical Power



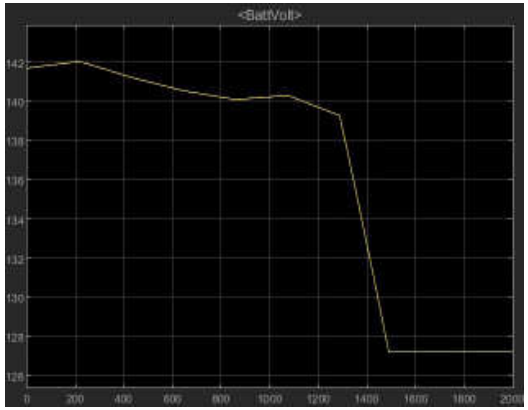
(c)SOC Of Battery



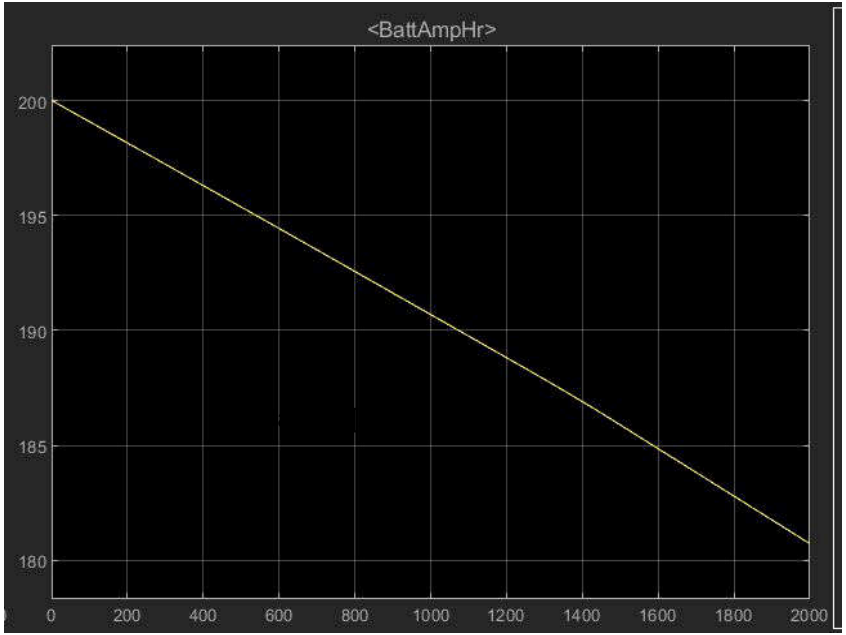
(d)Battery Power



(e)Battery Current



(f)Battery Voltage



(g) Battery [Ah]

Parameter	Value
Power	15 kW
Vehicle Velocity	43 km/h
Distance Traveled	24 km
Mechanical Power	17.5 kW
Net Electrical Power	19 kW
Battery Power	19 kW

Cell Modules

Selecting cells for an EV accumulator pack is crucial for optimizing performance, safety, and cost. Key factors include:-

- Energy Density : Affects driving range.
- Power Density : Impacts acceleration and regenerative braking.
- Lifecycle : Determines longevity.
- Safety : Prevents thermal runaway.
- Cost per kWh : Influences overall cost.
- Supply Chain Stability : Ensures availability.



The best cell which comes of our battery pack manufacturing comes out to be as **3.6V 50Ah NMC chemistry Prismatic cell**, which is manufactured by **Hawkeye Innovations, LLC**

Specification	Value
Nominal Voltage	3.6V
Capacity	50Ah
Energy Density	200-250 Wh/kg
Thermal Stability	Up to 60°C (operating)
Cycle Life	2000-3000 cycles

Accumulator Pack Configuration

To configure a 120V, 200Ah battery pack using 3.6V, 50Ah NMC cells:

- Connect 34 cells in series to achieve approximately 122.4V.
- Replicate this series string four times in parallel to meet the 200Ah capacity.

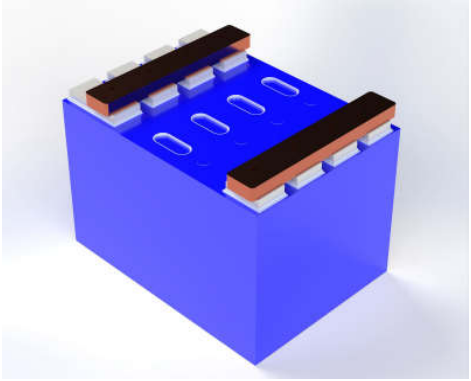
This design uses 144 cells in total, enhancing reliability and longevity. The table below summarizes the pack configuration:

Parameter	Value
Cell Voltage	3.6V
Cell Capacity	50Ah
Series Cells	36 cells
Parallel Strings	4 strings
Pack Voltage	~129.6V
Pack Capacity	200Ah
Total Cells Used	144 cells

Cell segments

This outlines the configuration of cell segments for a 200Ah, 120V battery pack using 3.6V, 50Ah NMC cells.

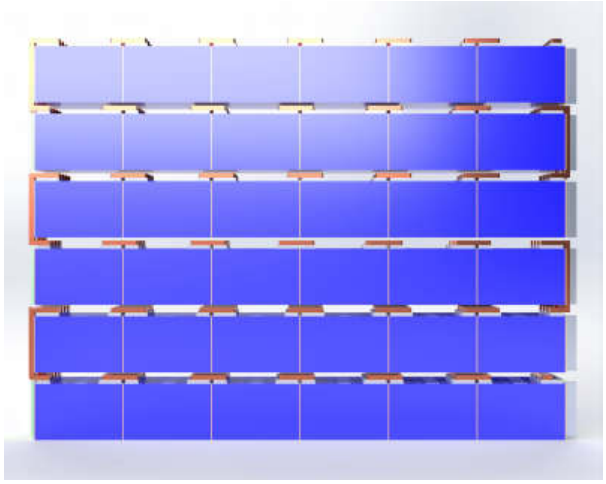
Parameter	Value
Voltage	120V
Capacity	200Ah
Cells	144 x 3.6V, 50Ah NMC cells
Series Cells per Segment	36 cells
Parallel Segments	4 segments
Total Cells Used	144 cells
Module Design	Four modules, each containing 36 cells in series

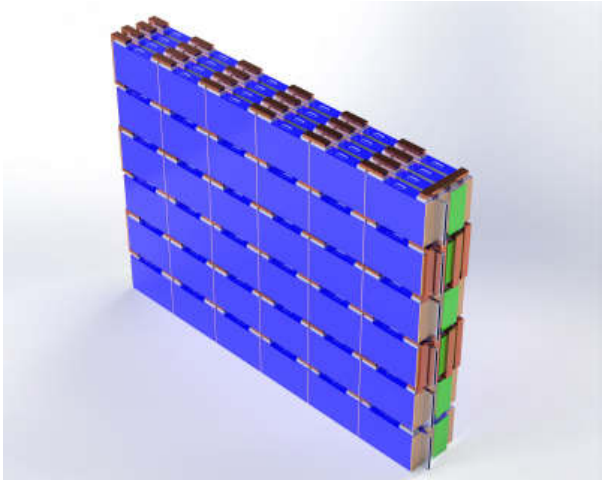


A part of cell segment

Cell Module Packaging & Interconnect Design

- Connect 36 cells in series, each composed of 4 parallel cells, to achieve a total voltage of 129.6V with a capacity of 200Ah.
- Use robust, heat-resistant materials and cooling systems for thermal management.
- Employ high-quality conductive materials like copper for busbars and ensure secure electrical connections.
- Integrate a BMS for monitoring cell parameters and active balancing.
- Implement protection mechanisms like fuses and mechanical supports to safeguard against faults and physical damage.





Busbar Calculations:

Parameter	Value
Maximum current	600A
Internal resistance	0.0008 ohms
Maximum temperature	50°C
Operating temperature	25°C
Thermal conductivity of copper	401 W/m·K (at 25°C)
Distance from cell to cooling system (LLL)	0.05 meters (50 mm or 5 cm)
Rth	0.001054m2·K/W

A =1.796×10−3m2

Hence, **7.6 mm thickness, 18 mm width busbar** used.

Material - copper

Heat Calculation:

Parameter	Value
Internal resistance of cell (R)	0.8 mohm
Current at 3C	150 A
No. of cells	144
Cross-sectional area of a single cell (AcellA_{cell}Acell)	0.04216546m2
Temperature difference (ΔT)	25◦C

h = 18.30

Hence, the battery pack must be equipped with the liquid cooling method to smoothly operate and maintain its life cycles and health.

Battery Thermal Analysis And Casing

Battery Thermal Analysis

Accumulator Specifications :

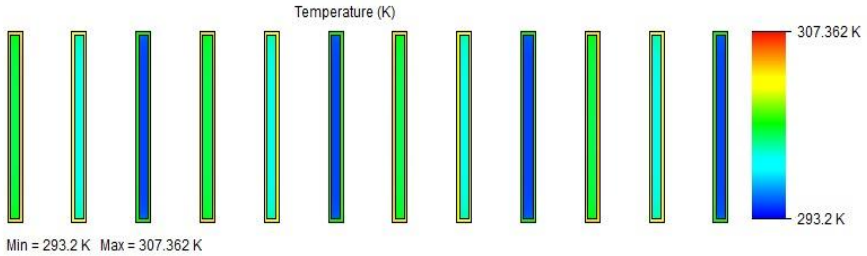
Parameter	Value
Type	Lithium-ion NMC Prismatic Cell
Cell Voltage	3.65V
Cell Capacity	50Ah
Configuration	36S4P
Maximum Discharge Rate	600A

Thermal Management Requirements

Parameter	Value
Peak Heat Generation	259.2 kW (during peak load)
Desired Temperature Range	20°C-25°C
Cooling Medium	Liquid (water/glycol mix)
Heat Transfer Coefficient	500 $W/m^2 \cdot K$

Flow Rate:

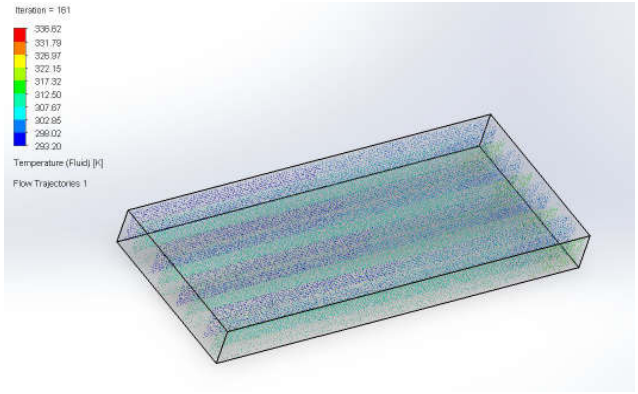
Parameter	Value
Mass Flow Rate (m\dot{m})	2.479 kg/s
Density of Coolant (ρ\rho)	1000 kg/m^3
Volumetric Flow Rate	2.479 L/s



Side View Thermal Gradient

Cooling Liquid Flow Rate:

Parameter	Value
Heat Transfer Rate (Q)	259,200 W or 259.2 kW
Specific Heat Capacity of Coolant (c_p)	4.18 kJ/kg·°C
Temperature Difference (ΔT\Delta TΔT)	25°C
Mass Flow Rate (m˙\dot{m}m˙)	2.479 kg/s



Battery Thermal Fluid Flow

Accumulator Casing

Material Selected : Aluminum Alloy 6061

Criteria	Requirement	Al 6061 Performance
Mechanical Strength	High tensile and yield strength	Tensile Strength: 310 MPa Yield Strength: 276 MPa
Weight	Lightweight	Density: 2.70 g/cm^3
Thermal Conductivity	Good heat dissipation	Thermal Conductivity: 167 W/m·K
Corrosion Resistance	High resistance to corrosion	Excellent corrosion resistance
Manufacturability	Easy to machine and form	Good machinability and weldability
Cost	Cost-effective	Moderate cost



Electrical Isolation Safety for Accumulator Casing

Materials Selected:

- Mica Sheets: For insulation plates
- FS1000: For insulation tapes

Property	Mica Sheets	FS1000 Insulation Tape	Justification
Dielectric Strength	1180 kV/cm	600 kV/cm	Prevents electrical shorts and ensures reliable insulation.
Thermal Stability	Up to 1000°C	Up to 150°C	Maintains performance under high temperatures.
Mechanical Strength	Tensile Strength: 210 MPa	High tensile strength and durability	Provides mechanical protection from impacts and vibrations.
Flame Retardancy	Excellent (inherent property)	UL 94 V-0 rating	Enhances safety by preventing fire propagation.
Flexibility	Rigid	Flexible	Facilitates easy application and coverage.
Density	2.7 g/cm^3	1.5 g/cm^3	Lightweight, adding minimal weight to the overall structure.

Battery Management System

Battery Management System

The Orion BMS (Battery Management System) is an advanced and comprehensive solution designed to protect and optimize the performance of lithium-ion battery packs. Here are several key reasons why using the Orion BMS is beneficial:

Parameter	Specification
Cell Capacity	Up to 180 cells in series
Cell Voltage Range	0 to 5V per cell
Total Pack Voltage Range	Up to 900V
Current Measurement Range	-1000A to +1000A
Temperature Measurement Range	-40°C to 125°C
Communication Protocols	CANBUS 2.0B, optional analog outputs
Power Consumption	Typical < 2W
Safety Features	Over-voltage, under-voltage, over-current, over-temperature protections, fault detection, and isolation monitoring

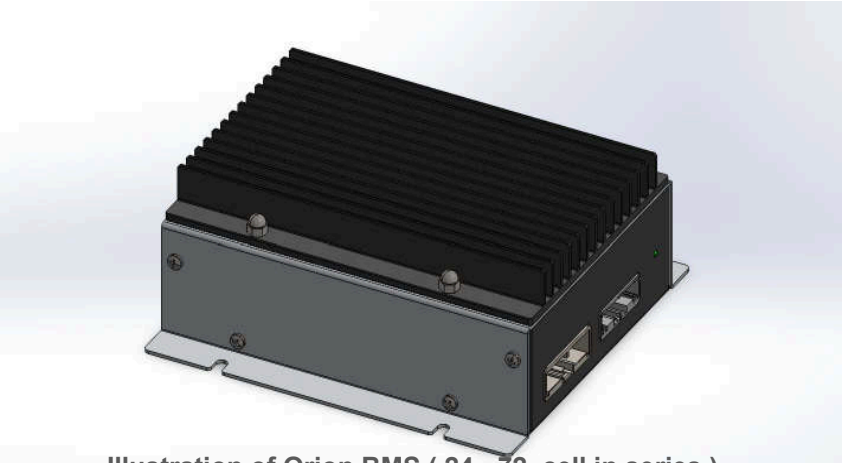
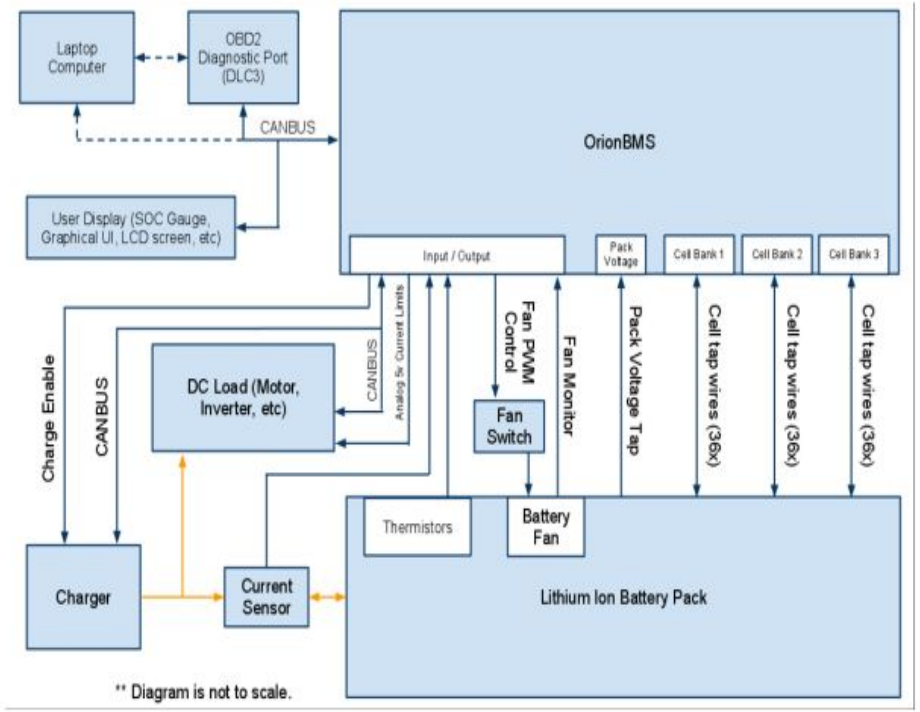


Illustration of Orion BMS (24 - 72 cell in series)

Wiring Overview of Orion BMS 2

- ➔ OrionBMS : Central to the system, managing inputs/outputs, pack voltage, and cell bank monitoring.
- ➔ Components Connected:
  - ◆ Laptop Computer: Communicates with Orion BMS via OBD2 Diagnostic Port (CANBUS).
  - ◆ User Display: Shows SOC gauge, graphical UI, etc., connected via CAN BUS.
  - ◆ Charger: Enables charge through CANBUS.
  - ◆ DC Load: Includes motor and inverter, connected via CAN BUS.
  - ◆ Thermistors and Fan: Monitored and controlled by Orion BMS.
  - ◆ Current Sensor: Monitors current flow to/from the battery.



Overview of system connections

CAN Interfaces Of Orion BMS

Use twisted pair wires, shielded cables, and proper grounding; terminate with 120-ohm resistors at each end; support various baud rates (125, 250, 500, or 1000 Kbps) on CAN1 and CAN2 interfaces.

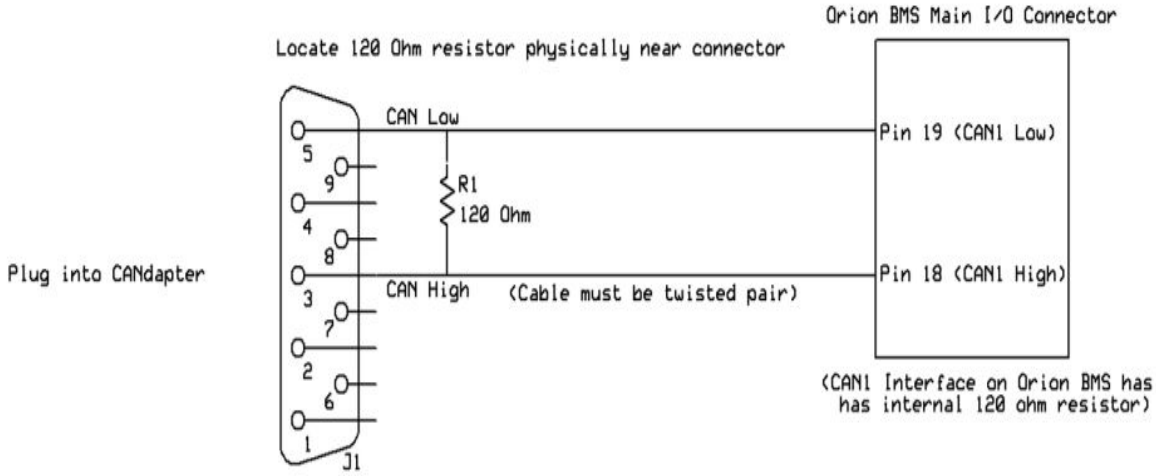
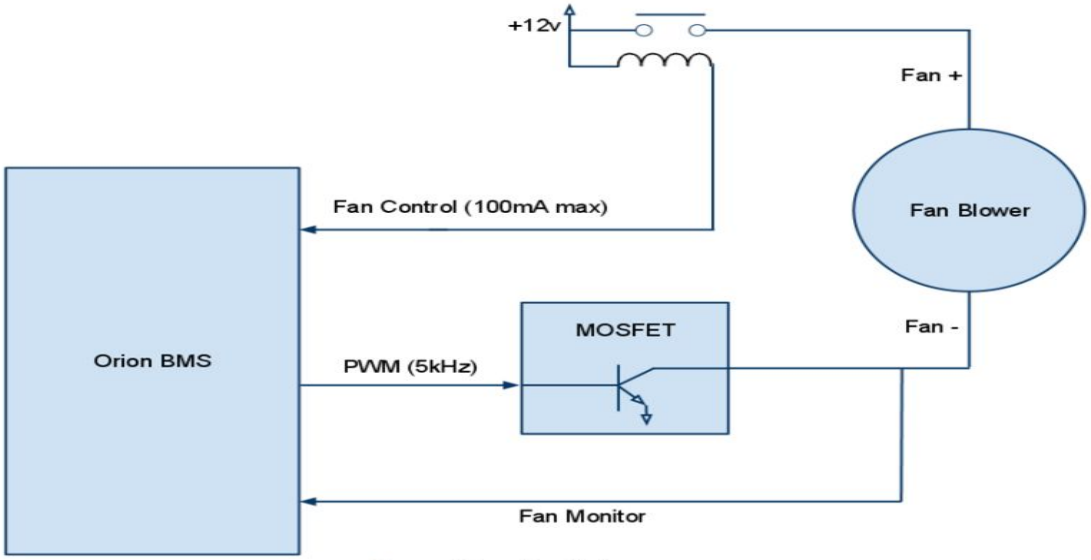


Diagram for connecting the CAN1 interface directly into a DB9 to be connected to the CANdapter

Fan Controller

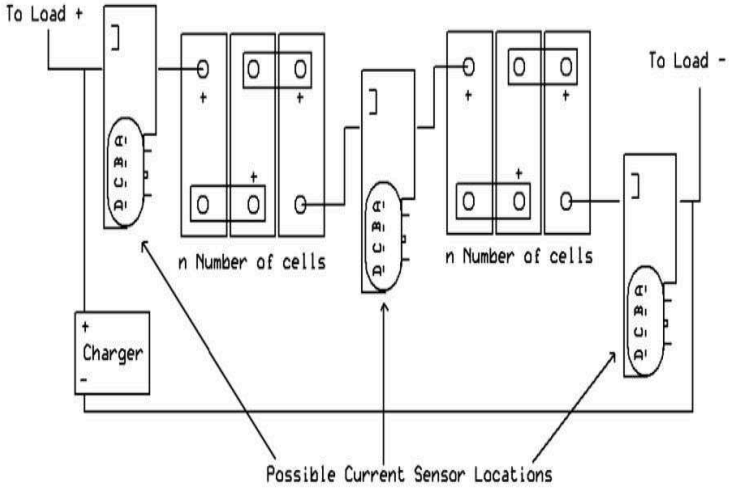
The Orion BMS provides robust thermal management with support for up to 4 thermistors expandable to over 800, fan control options including PWM for variable speed, and fan monitoring for voltage detection. Essential for maximizing battery performance and lifespan.



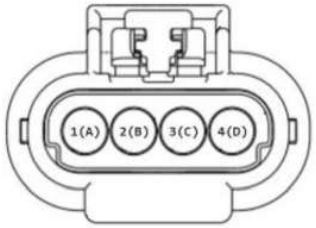
General fan circuit diagram.

Current Sensor and Thermistor Setup

The Orion BMS ensures precise current sensing and temperature monitoring crucial for efficient battery management. With flexible wiring options for current sensors and thermistors, it guarantees accuracy and safety. Strategic placement of thermistors throughout the battery pack, including an option for intake air temperature monitoring, enhances cooling management. Expansion modules allow for monitoring up to 804 thermistors for comprehensive temperature tracking.



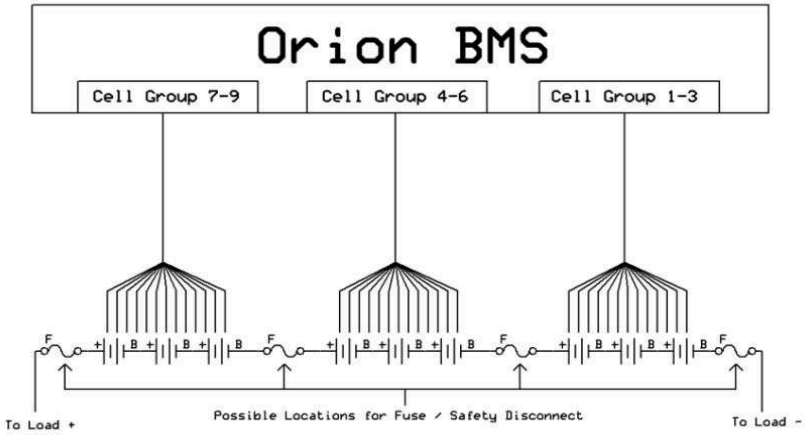
Current Sensor Connector Diagram



Pin	Description
1 (A)	+5V Current Sensor Power
2 (B)	Current High Channel
3 (C)	Current Sensor Ground
4 (D)	Current Low Channel

Safety Disconnects and Fuse:

Mid-pack disconnects ensure safety in high voltage or high amperage packs. Orion BMS provides 2.5kV safety isolation, emphasizing proper placement to prevent damage. Flexible wiring placement and diagrams aid in installation.

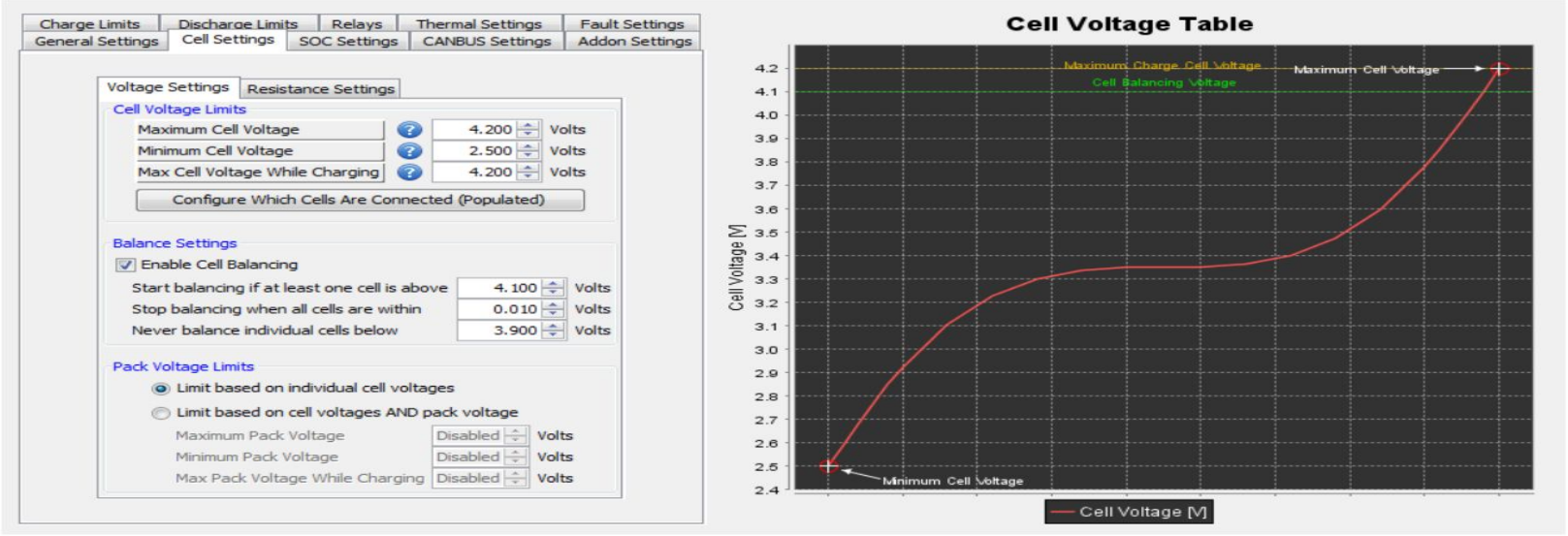


Voltage Tap Harness Lengths:

Available in 6-foot and 12-foot options. Length affects accuracy; revisions B, C, and D recommend up to 12 feet, while revision E supports up to 30 feet for optimal accuracy.

Firmware

Orion BMS 2 firmware ensures battery safety with comprehensive fault detection and response. It monitors voltage, temperature, current, and isolation, triggering alerts, throttling, shutdowns, and logging. Customizable settings via Utility software provide tailored protection, while updates continuously improve safety and performance.



Cell Settings

The Orion BMS firmware controls hardware functions and undergoes a straightforward update process involving preparation, downloading, connecting to Utility O2, installing, and verifying. Detailed instructions ensure optimal performance and safety.



Live Data Handling

Accumulator Isolation Relay (AIR)



An AIR is a normally open contactor that is regulated by the vehicle's shutdown circuit. The AIRs are meant to isolate both poles of the tractive battery, ensuring that no high voltage is present on the accumulator output.

Description	Attribute
Coil Voltage	24 VDC
Contact current rating	350 A
Switching voltage	800 VDC
Coil Current	97 mA
Operating temperatures	Max - +85 C Min - -55 C

IMD and Safety

Insulation Monitoring Device (IMD)

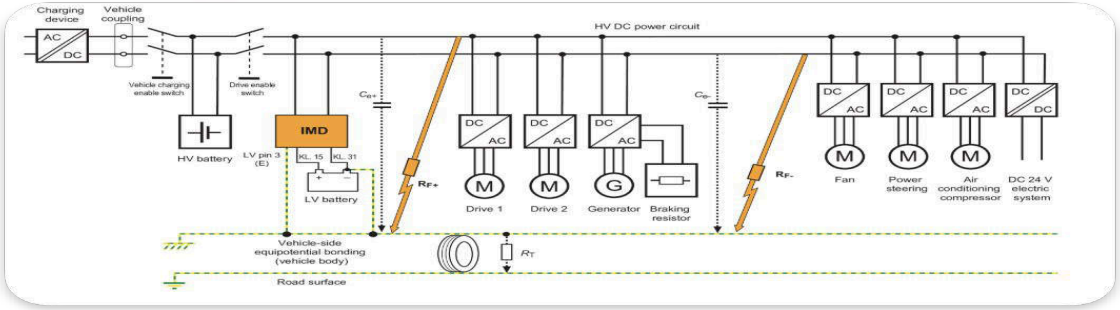
The ISOMETER® iso175

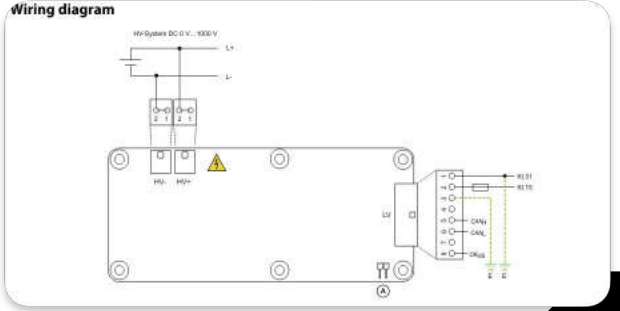
An insulation monitoring device (IMD) designed to ensure the safety and reliability of electrical systems, particularly in DC systems up to 1000 V. Its operating principle involves connecting between live supply conductors and earth, applying a measuring voltage, and detecting faults by measuring resulting current



Key features

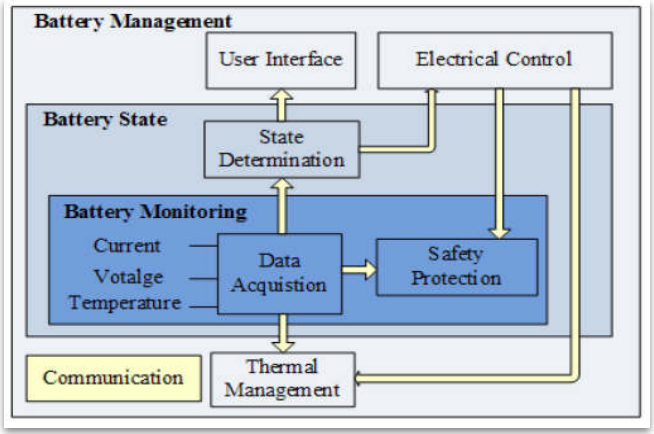
include compatibility with 12V and 24V DC systems, continuous insulation resistance measurement, rapid response time for fault detection, offline self-testing for HV connection monitoring, continuous earth connection monitoring, and undervoltage detection. It also offers load-dump protection and can measure insulation for system leakage capacitances up to 10 µF.

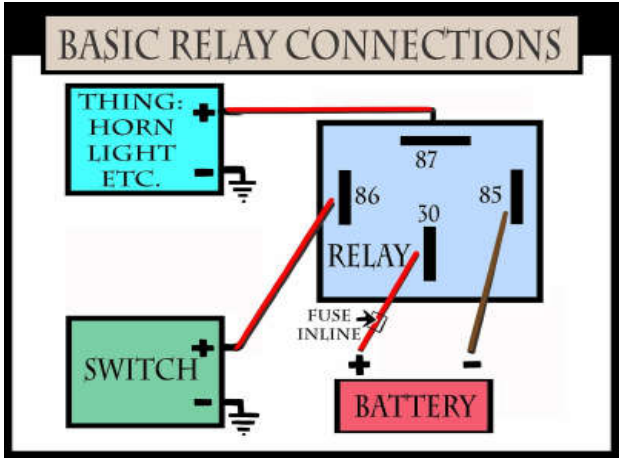




Design For Safety

- **Cells Body and Battery Pack:** Robust Packaging arrangement, monitoring cell current, voltages,states etc and shutdown button for malfunctioning are all installed in vehicle.
- **High Voltage System Safety:** Insulation and grounding, Clearly marked high-voltage components.
- **Electrical Insulation and Fusing:** Adequate spacing and insulation is provided. Also different kinds of relays (AIR, SSR etc) are used.
- **Emergency Shutdown System:** Shutdown mechanism that disconnect the battery, easily accessible to the driver





Battery Auxiliaries

Battery Auxiliaries

Fuses:
Bussmann Eaton HV Fuse**
- Purpose : Overcurrent protection.
- Specs : 250A, 150V DC.
- Reasons : Suitable for 200Ah pack, exceeds voltage requirements, reliable, safety compliant.
- Install : Near positive terminal, secure, low-resistance connections.

Maintenance Plugs:
TE Connectivity HVP 800**
- Purpose : Safe disconnection for maintenance/emergencies.
- Specs : 250A, up to 450V.
- Reasons : Durable, safe, prevents accidental disconnections.
- Install : Accessible locations, proper insulation, secure connections.

HV/LV Connectors
- Purpose : HV for high voltage, LV for lower voltage applications.
- Specs : 200A, 150V DC.
- Reasons : Suitable ratings, high-quality insulation, EMI shielding.
- Install : Secure HV connectors, accessible LV connectors.

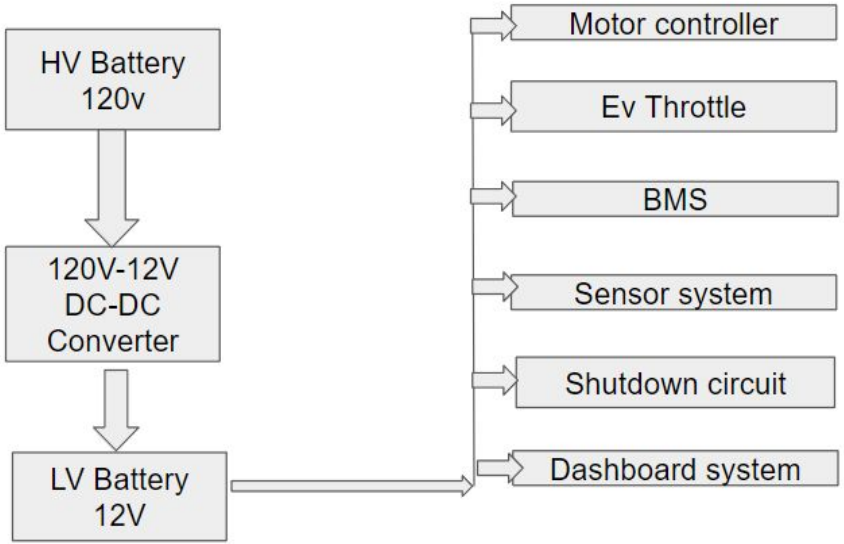
High Voltage Cabling:
Alpha Wire Xtra-Guard 4
- Specs : 2/0 AWG, 600V.
- Reasons : Handles high current, high-quality insulation, EMI shielding.
- Install : Avoid sharp bends, secure with clamps, organize LV cables properly.

IIITDMJ racing

PI-EV

Electrical Systems

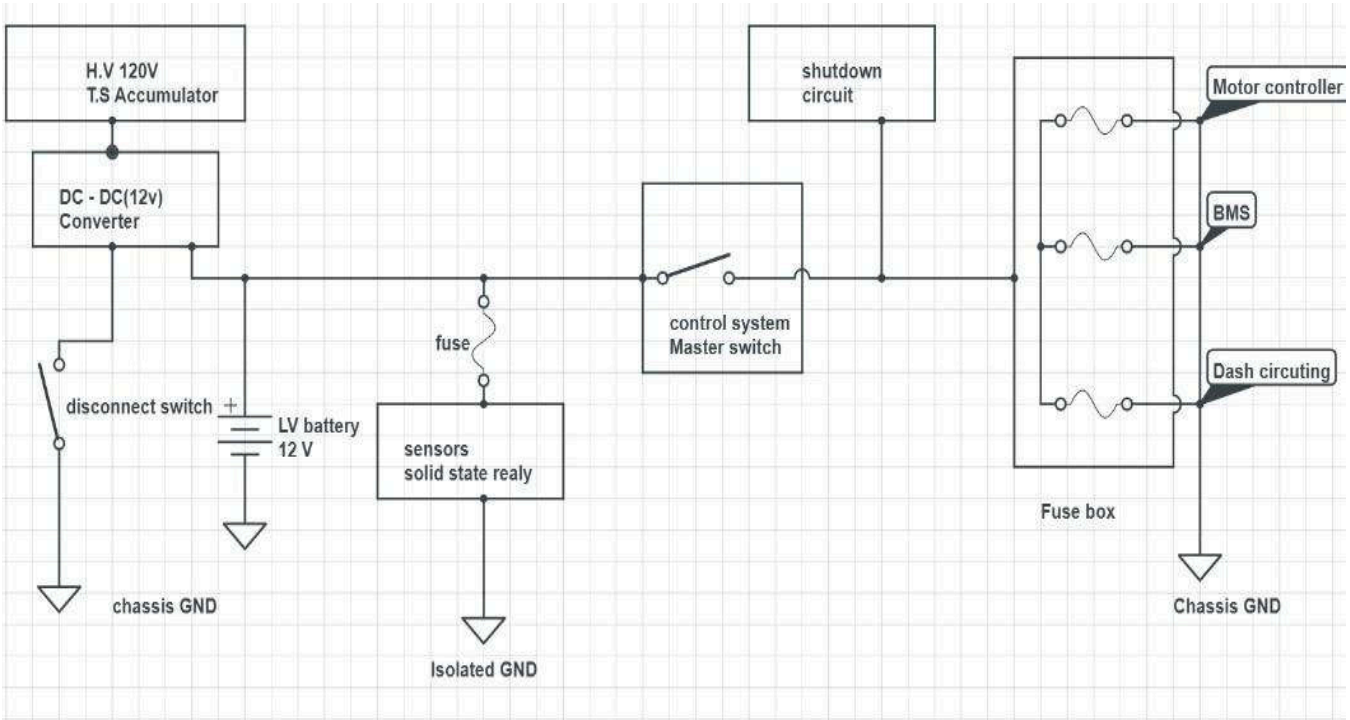
Low Voltage System Architecture



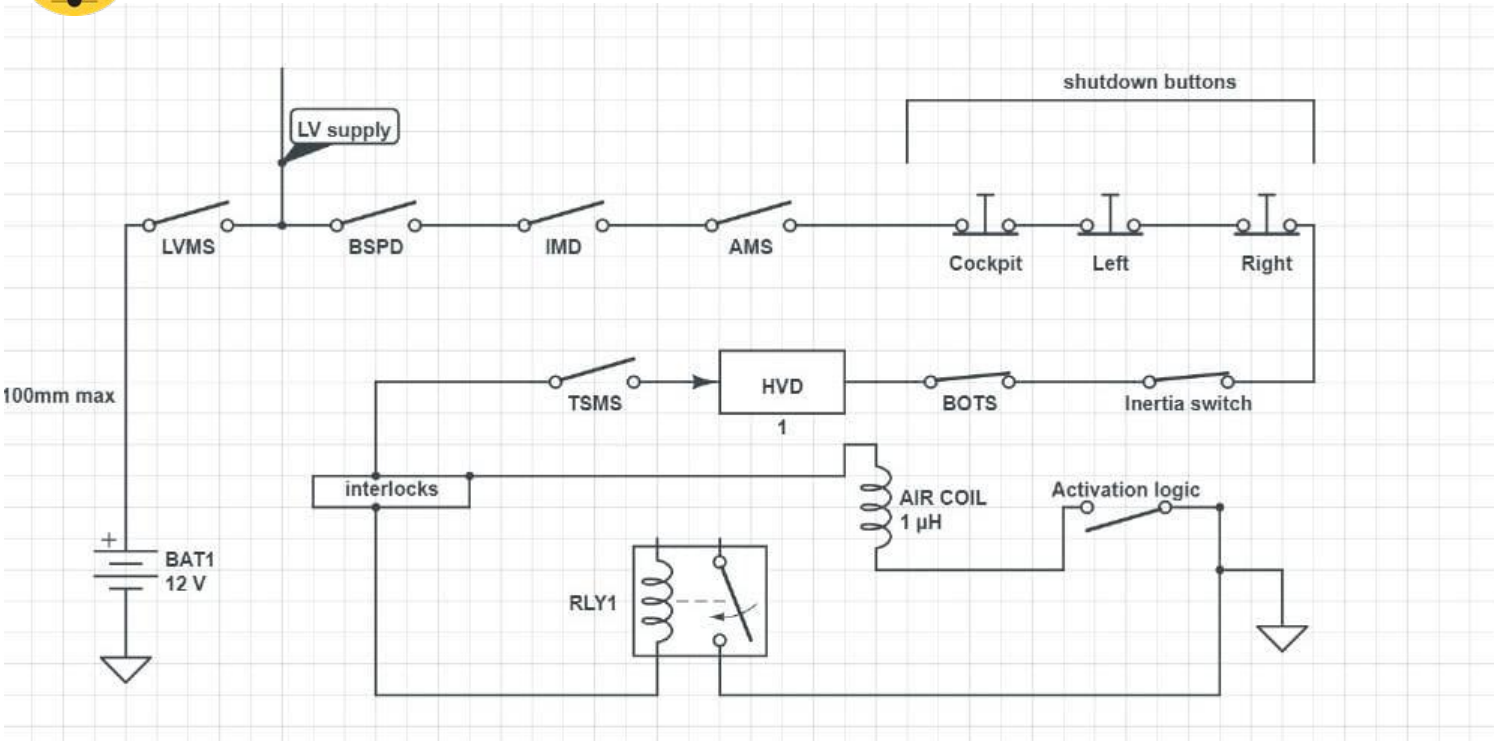
Connect cells in series and parallel for desired voltage and capacity. Utilize robust materials, cooling systems, high-quality conductors, BMS for monitoring, and protection mechanisms.

LV System architecture

Circuit diagram:



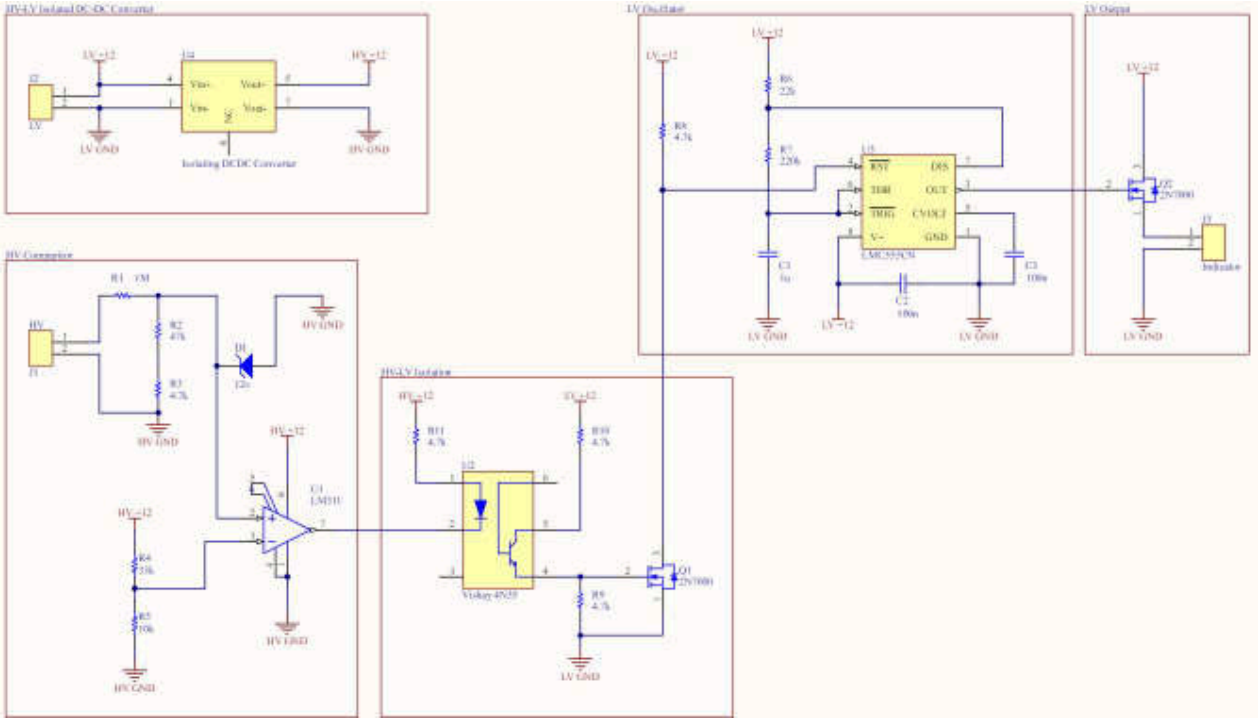
Shutdown Circuit



The shutdown circuit is a meticulously designed safety system, ensuring swift power disconnection during emergencies. Through components like LVMS, BSPD, IMD, AMS, physical switches, inertia switch, BOT switch, HVD, and TSMS, it guarantees comprehensive protection.

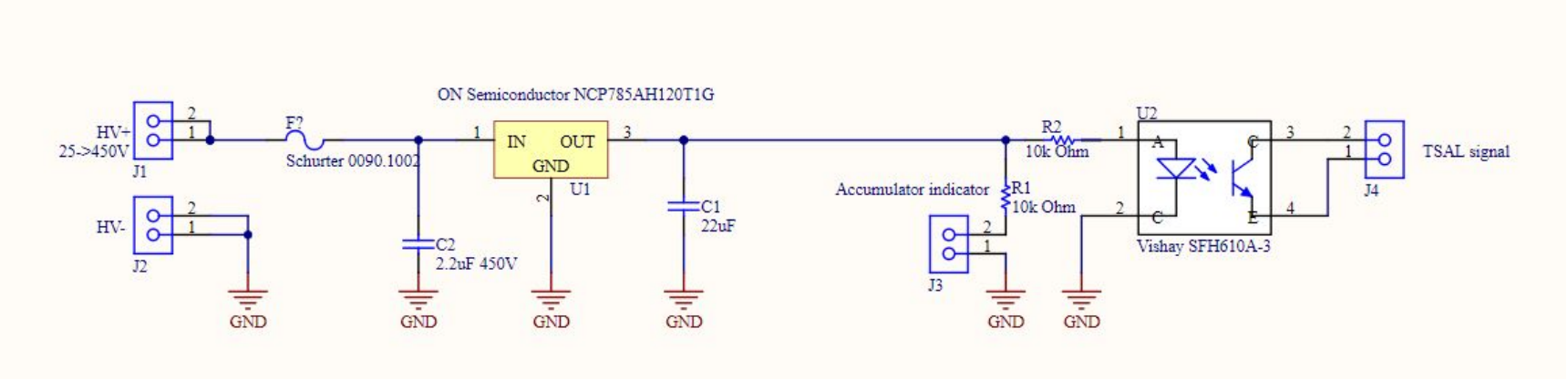
Tractive System Active Light (TSAL)

The Tractive System voltage level is indicated by the Tractive System Active Light (TSAL), powered by the Ground Low Voltage System. Below 60 V DC, it remains green; above, it flashes red (2-5 Hz).



Accumulator Indicator:

It shows if more than 60 VDC exist behind the AIRs. It gets the connection from HV and interprets if the voltage exceeds.



Hardware Packaging, Harness, and Connectors:

- Essential for reliable and safe operation.
- Ensure secure connections and effective communication.
- Monitors voltage exceeding 60 VDC behind the AIRs.

High Voltage (HV) Circuits:

- Shielded EV cables connect HV components.
- Rated temperature: -40℃ to +125℃.
- Compliant with safety standards.

Low Voltage (LV) Circuits:

- Cable specification: 10 19/23 BC TXL per J1128 ESB-M1L123-A MS-8288.

Hardware Packaging and Waterproofing:

- Raychem ES Caps for secure wire termination.
- DR-25 Heat Shrink provides insulation and protection.

This overview ensures all wiring and connection components are secure, insulated, and compliant with safety standards.

Data Logging Methods, Driver Interfaces, Telemetry

Component	Type	Location	Data Rate/Accuracy
Speed Sensor	Hall Effect Sensor	Wheel Hubs	-
Temperature Sensor	NTC Thermistor	Battery Pack	±0.5°C
Voltage Sensor	Voltage Divider Circuit	Battery Pack	±0.1%
Current Sensor	Hall Effect Sensor	Battery Pack	±0.1%
Accelerometer	MEMS Accelerometer	Chassis	±0.01g
Digital Display	LCD Screen	Dashboard	1 Mbps
Telemetry Transmitter	Wireless Communication Device	Chassis	1 Mbps
Emergency Stop	Mechanical Button	Cockpit	-

The integration of robust driver interfaces, data logging systems, and telemetry enhances the performance, safety, and competitiveness of Formula Student Electric Vehicles. By prioritizing design considerations, data usage, integration, and regulatory compliance, teams can develop comprehensive systems that meet the demands of the FSAE competition and push the boundaries of electric vehicle technology.